

IEOR4004 - Assignment4

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Code Link: https://drive.google.com/file/d/1LID0Is-Ga6_v97PnNLaOpALb_y2Nakx7/view?usp=sharing

Map:

[SP brute-force]

- restaurant: Restaurant1
- stop order: ['Hotel', 'GreenwichVillage', 'Restaurant1', 'MetMuseum', 'Columbia']
- total distance: 23800.0
- expanded route nodes: ['Hotel', 'E', 'G', 'GreenwichVillage', 'K', 'H', 'F', 'Restaurant1', 'F', 'MetMuseum', 'C', 'Columbia']

Assignment4 SP Route (open path)



Description:

I enumerate (1) which restaurant to go to and (2) all possible visiting orders of the three landmarks plus that restaurant.

For any candidate order, I compute the actual route between each consecutive pair of "stops" using the **shortest path** on the graph (so each segment between stops uses the cheapest path through intermediate intersections). I then sum the shortest-path lengths for all consecutive legs and select the solution with the **minimum total travel distance**.

For the one-day plan, after completing the final required location, **the itinerary can naturally end with continuing to the next destination (e.g., the airport)**, that's why I am using the SP rather than TSP method.

p.s. if a "back to start" route is needed for presentation, returning to **Hotel** can be shown as the closing leg.

which results in:

- restaurant: Restaurant1

- stop cycle: ['Hotel', 'GreenwichVillage', 'Columbia', 'MetMuseum', 'Restaurant1', 'Hotel']
- total distance: 33400
- expanded route nodes: ['Hotel', 'E', 'G', 'GreenwichVillage', 'G', 'E', 'C', 'Columbia', 'C', 'MetMuseum', 'F', 'Restaurant1', 'F', 'Hotel']